6. Verify the additive and multiplicative inverse property for the rational numbers $\frac{-7}{17}$ and $\frac{17}{27}$.

## Objective Type Questions

7. Closure property is not true for division of rational numbers because of the number

(A) 1 (B) –1 (C) 0 (D) $\frac{1}{2}$

8. $\frac{1}{2}-\left(\frac{3}{4}-\frac{5}{6}\right)\neq\left(\frac{1}{2}-\frac{3}{4}\right)-\frac{5}{6}$ illustrates that subtraction does not satisfy the _____ property for rational numbers.

(A) commutative (B) closure (C) distributive (D) associative

9. Which of the following illustrates the inverse property for addition?

(A) $\frac{1}{8}-\frac{1}{8}=0$ (B) $\frac{1}{8}+\frac{1}{8}=\frac{1}{4}$ (C) $\frac{1}{8}+0=\frac{1}{8}$ (D) $\frac{1}{8}-0=\frac{1}{8}$

10. $\frac{3}{4}\times\left(\frac{1}{2}-\frac{1}{4}\right)=\frac{3}{4}\times\frac{1}{2}-\frac{3}{4}\times\frac{1}{4}$ illustrates that multiplication is distributive over

(A) addition (B) subtraction (C) multiplication (D) division




We know that different operations with the same pair of rational numbers usually give different answers. Check the following calculations which are some interesting exceptions in rational numbers.

(i) $\frac{13}{4}+\frac{13}{9}=\frac{13}{4}\times\frac{13}{9}$ (ii) $\frac{169}{30}+\frac{13}{15}=\frac{169}{30}\div\frac{13}{15}$

Amazing …! Isn't it? Try a few more like these, if possible.

**Think**



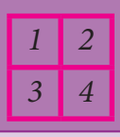
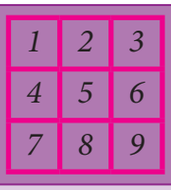
Observe that, $\frac{1}{1.2}+\frac{1}{2.3}=\frac{2}{3}$

$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}=\frac{3}{4}$

$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\frac{1}{4.5}=\frac{4}{5}$

Use your reasoning skills, to find the sum of the first 7 numbers in the pattern given above.

## 1.6 Introduction to Square Numbers

| | | |
|---|---|---|
|  |  |  |
| This is a square of side 1 unit. It is 1 squared. We write this as $1^2$ | This is a square of side 2 units. It is 2 squared. We write this as $2^2$ | This is a square of side 3 units. It is 3 squared. We write this as $3^2$ |
| $1^2 = 1 \times 1 = 1$ | $2^2 = 2 \times 2 = 4$ | $3^2 = 3 \times 3 = 9$ |



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